

C The detailed simulation setup

This paper uses five different applications of nonlinear KFs to test the effectiveness of the new framework: 3D target tracking, terrain-referenced navigation, synchronous generator state estimation, pendulum state estimation, and battery state estimation.

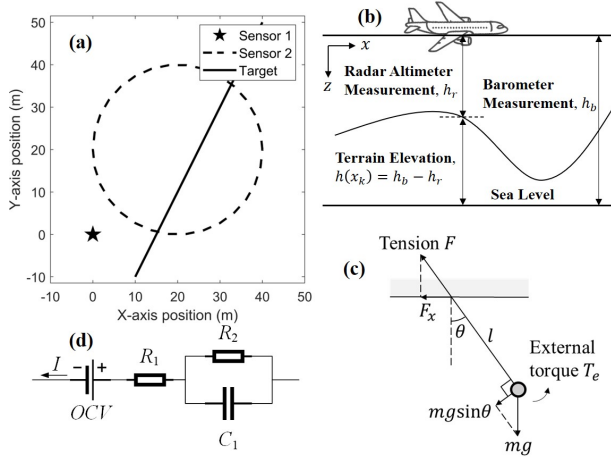


Fig. C.1. Examples of nonlinear systems used in this paper for validation. (a) 3D object tracking. (b) Terrain-referenced navigation. (c) Pendulum state estimation. (d) Battery state estimation.

C.1 3D target tracking

This example is adapted from Example 3.3.1 in [39]. As shown in Fig. C.1(a), a target moves in 3D space, and two sensors are used to track its location. The first sensor is at the origin, and the second sensor moves in a circle. The second sensor's location (x_{s1}, x_{s2}, x_{s3}) at step $k = 1, 2, \dots, 30$ is represented by

$$\begin{cases} x_{s1,k} = 20 + 20 \cos\left(\frac{(k-1)\pi}{15}\right), \\ x_{s2,k} = 20 + 20 \sin\left(\frac{(k-1)\pi}{15}\right), \\ x_{s3,k} = 0. \end{cases} \quad (C.1)$$

The states are the target's positions (x_1, x_2, x_3) and speeds (v_1, v_2, v_3) along the three axes. The measurements z_1, z_2 are the distances between the target and the two sensors. The inputs are the target's accelerations along the three axes, which are assumed to be zero in this example. When the time interval is set to $\Delta t = 1s$, the system's state transition and measurement functions can be written as (C.2) and (C.3). Note that the noises

are omitted in the equations.

$$\begin{bmatrix} x_{1,k} \\ x_{2,k} \\ x_{3,k} \\ v_{1,k} \\ v_{2,k} \\ v_{3,k} \end{bmatrix} = \begin{bmatrix} x_{1,k-1} + v_{1,k-1} \\ x_{2,k-1} + v_{2,k-1} \\ x_{3,k-1} + v_{3,k-1} \\ v_{1,k-1} \\ v_{2,k-1} \\ v_{3,k-1} \end{bmatrix}, \quad (C.2)$$

$$\begin{bmatrix} z_{1,k} \\ z_{2,k} \end{bmatrix} = \begin{bmatrix} \sqrt{\sum_{i=1}^3 x_{i,k}^2} \\ \sqrt{\sum_{i=1}^3 (x_{i,k} - x_{si,k})^2} \end{bmatrix}. \quad (C.3)$$

The true initial states of the target are $x_{1,0} = 10, x_{2,0} = -10, x_{3,0} = 50, v_{1,0} = 1, v_{2,0} = 2, v_{3,0} = 0$; the initial state covariance matrix P_0 is diagonal, and its diagonal elements are 100, 100, 100, 0.01, 0.01, and 0.01. The initial state estimations are generated randomly from the multivariate Gaussian distribution with covariance P_0 and mean equal to the true initial states. The process noise covariance matrix is also diagonal, and its diagonal values are 0, 0, 0, 10^{-6} , 10^{-6} , and 10^{-6} .

C.2 Terrain-referenced navigation

This example is adapted from Example 3.3.2 in [39]. As shown in Fig. C.1(b), a plane flies above an uneven landscape. The states are the planes' positions along the two axes, denoted as x_1 and x_2 . The measurement z_k is the terrain's elevation at the plane's present location, which, in practice, is calculated as the difference between the barometer and radar altimeter measurements. The plane knows the contour map in advance, so the terrain's elevation can tell the aircraft its possible locations. However, this task is difficult because all the points on a contour line have the same elevation, and system knowledge must be combined for accurate state estimations. The system inputs are the plane's speeds along the x and y axes, which are 0.5 km/s and 0, respectively. When the time interval is set to $\Delta t = 1s$, the system's state transition and measurement functions can be written as (C.4) and (C.5). Note that all the units for distance in this example are kilometers, and the noises are omitted in the equations.

$$\begin{bmatrix} x_{1,k} \\ x_{2,k} \end{bmatrix} = \begin{bmatrix} x_{1,k-1} + 0.5 \\ x_{2,k-1} \end{bmatrix}, \quad (C.4)$$

$$z_k = \sin \sqrt{\left(\frac{x_{1,k}}{40}\right)^2 + \left(\frac{x_{2,k}}{40}\right)^2}. \quad (C.5)$$

The true initial states of the target are $x_{1,0} = 10$ km, $x_{2,0} = 10$ km; the initial state covariance matrix P_0 is diagonal, and its diagonal elements are both 1 km². The initial state estimations are generated randomly from the multivariate Gaussian distribution with covariance

P_0 and mean equal to the true initial states. The process noise covariance matrix is also diagonal, and its diagonal values are both 0.25 m^2 .

C.3 Synchronous generator state estimation

This example is adapted from a research paper [40]. The system model is precisely the same as in the paper. The states of the system are the rotor angle δ , rotor speed $\Delta\omega$, the q-axis component of the voltage e'_q , and the d-axis component of the voltage e'_d . For simplicity, these states are denoted as x_1 to x_4 . The measurement z_k is electrical output power. The system's inputs include the mechanical input torque T_m , the steady-state internal voltage of the armature E_{fd} , and the terminal bus voltage V_t . For simplicity, these inputs are denoted as u_1 to u_3 . The time interval is set to $\Delta t = 0.1 \text{ ms}$. The input profiles are the same as in the paper, which are:

$$\begin{cases} u_{1,k} = 0.8, \\ u_{2,k} = 2.11 + 0.0002k, \\ u_{3,k} = 1.002. \end{cases} \quad (\text{C.6})$$

All the model parameters are selected to be the same as in [40]. The system's state transition and measurement functions can be written as (C.7) and (C.8). Note that the noises are omitted from the equations.

$$\begin{bmatrix} x_{1,k} \\ x_{2,k} \\ x_{3,k} \\ x_{4,k} \end{bmatrix} = \begin{bmatrix} x_{1,k-1} \\ x_{2,k-1} \\ x_{3,k-1} \\ x_{4,k-1} \end{bmatrix} + \begin{bmatrix} \frac{377x_{2,k-1}\Delta t}{13} \\ \frac{\Delta t}{13} \left[u_{1,k-1} - \frac{u_{3,k-1}x_{3,k-1}\sin x_{1,k-1}}{0.375} \right] \\ \frac{\Delta t}{0.131} [u_{2,k-1} - x_{3,k-1}] \\ \frac{-x_{4,k-1}\Delta t}{0.0131} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\Delta t}{13} [0.9215u_{3,k-1}^2 \sin(2x_{1,k-1}) - 0.05x_{2,k-1}] \\ \frac{-4.4933\Delta t}{0.131} (x_{3,k-1} - u_{3,k-1} \cos x_{1,k-1}) \\ \frac{0.6911u_{3,k-1}\sin x_{1,k-1}\Delta t}{0.0131} \end{bmatrix}, \quad (\text{C.7})$$

$$z_k = \frac{u_{3,k-1}x_{3,k}\sin x_{1,k}}{0.375} + 0.9215u_{3,k-1}^2 \sin(2x_{1,k}). \quad (\text{C.8})$$

The true initial states of the system are $x_{1,0} = 0.4$, $x_{2,0} = x_{3,0} = x_{4,0} = 0$. The initial state covariance matrix P_0 is diagonal, and its diagonal elements are 10^{-4} , 10^{-10} , 10^{-4} , and 10^{-4} . The initial state estimations are generated randomly from the multivariate Gaussian distribution with covariance P_0 and mean equal to the true initial states. The process noise covariance matrix is also diagonal, and its diagonal values are 10^{-10} , 10^{-16} , 10^{-10} , and 10^{-10} .

C.4 Pendulum state estimation

As shown in Fig. C.1(c), one end of the pendulum is fixed to the ceiling with a thin rope. The system's states are the pendulum's angular position θ and angular speed

ω . The measurement z_k is the horizontal component of tension in the rope. The system input is the external torque exerted on the pendulum, which is zero in this example. The mass of the ball is $m = 1 \text{ kg}$, the length of the rope is $l = 1 \text{ m}$, and the gravity acceleration is $g = 9.8 \text{ m/s}^2$. The time interval is set to $\Delta t = 0.01 \text{ s}$. According to Newton's law, when the time interval is small, the system's state transition and measurement functions can be written as (C.9) and (C.10). Note that the noises are omitted in the equations.

$$\begin{bmatrix} \omega_k \\ \theta_k \end{bmatrix} = \begin{bmatrix} \omega_{k-1} - \frac{g}{l} \sin(\theta_{k-1})\Delta t \\ \theta_{k-1} + \omega_{k-1}\Delta t \end{bmatrix}, \quad (\text{C.9})$$

$$z_k = mg \cos \theta_k \sin \theta_k + ml\omega_k^2 \sin \theta_k. \quad (\text{C.10})$$

The true initial states of the system are $\omega_0 = 0$, $\theta_0 = \pi/4$. The initial state covariance matrix P_0 is diagonal, and its diagonal elements are both $(\pi/18)^2$. The initial state estimations are generated randomly from the multivariate Gaussian distribution with covariance P_0 and mean equal to the true initial states. The process noise covariance matrix is also diagonal; its diagonal values are 10^{-10} and 0 .

C.5 Battery state estimation

This example is adapted from a research paper [41]. As shown in Fig. C.1(d), a battery can be modeled as several circuit components, including a voltage source OCV (stands for the cell's open-circuit voltage), an internal resistor R_1 , a transfer resistor R_2 , and a capacitor C_1 . A battery's two most important states are its state of charge (SOC) and state of health (SOH). The SOC is the ratio of the battery's remaining capacity to its present maximum capacity, and the SOH is the ratio of the battery's present maximum capacity to its initial maximum capacity. In general, SOC describes the battery's energy level, while SOH describes the battery's aging level. The states of this system are the SOC, SOH, and capacitor voltage U_c . The measurement z_k is the battery's terminal voltage. The input is the charging current, which, in this example, is a three-level square wave shown below:

$$I_k = \begin{cases} -2, & 15 < k \leq 75, \\ 2, & 105 < k \leq 165, \\ 0, & \text{else.} \end{cases} \quad (\text{C.11})$$

The battery's initial maximum capacity is $Q_0 = 1 \text{ Ah}$. The parameters in the equivalent circuit model are chosen to be $R_1 = 0.01 \Omega$, $R_2 = 0.05 \Omega$, $1/(R_2C_1) = 0.008 \text{ s}$. The OCV is a function of SOC and SOH, and it can be written as:

$$OCV = \sum_{i=0}^9 \left(\frac{SOH - 0.8}{0.2} a_{100,10-i} + \frac{1 - SOH}{0.2} a_{80,10-i} \right) SOC^i, \quad (\text{C.12})$$

where $a_{100} = [1390.38, -6961.31, 14760.31, -17230.92, 12055.71, -5162.75, 1330.60, -196.37, 15.60, 2.96]$, $a_{80} = [813.94, -4229.96, 9345.49, -11415.38, 8396.15, -3801.07,$

1043.09, -165.29, 14.28, 2.96]. The time interval is set to $\Delta t = 1$ s, and the total simulation time is 180 seconds. The system's state transition and measurement functions can be written as (C.13, C.14). Note that the noises are omitted in the equations.

$$\begin{bmatrix} SOC_k \\ U_{c,k} \\ SOH_k \end{bmatrix} = \begin{bmatrix} SOC_{k-1} \\ U_{c,k-1} \exp\left(\frac{-\Delta t}{R_2 C_1}\right) \\ SOH_{k-1} \end{bmatrix} + \begin{bmatrix} I_{k-1} \Delta t \\ [1 - \exp\left(\frac{-\Delta t}{R_2 C_1}\right)] R_2 I_{k-1} \\ 0 \end{bmatrix} \quad (C.13)$$

$$z_k = OCV(SOC_k, SOH_k) + U_{c,k} + R_1 I_k \quad (C.14)$$

The true initial states of the system are $SOC_0 = 60\%$, $U_{c,0} = 0$, $SOH_0 = 90\%$. The initial state covariance matrix P_0 is diagonal; its diagonal elements are 0.04, 10^{-10} , and 0.01. The initial state estimations are 80%, 0, and 100%, respectively. The input current is noisy, and its standard deviation is 1 mA. This input noise is the only source of the process noises. Since the input current also appears in the measurement function, the actual measurement noise is the combination of the measurement noise of the voltmeter and the noise caused by the term $R_1 I_k$ in the measurement function.

D Supplementary results related to the “back out” step

Fig. 8, 9, D.1–D.3 compare the state estimation accuracy with and without the “back out” step. Additionally, Fig. D.4–D.8 quantify how often the proposed “back out” step is activated in practice by reporting the percentage of state updates that trigger this safeguard.

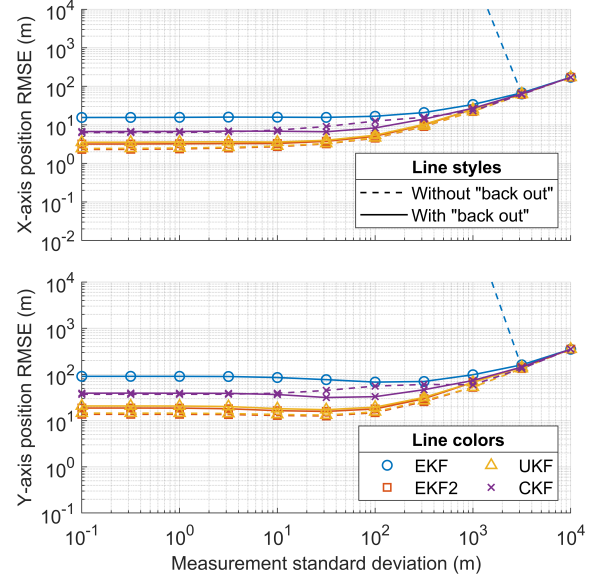


Fig. D.1. The root mean squared error of the state estimations with and without the “back out” step under different measurement noise setups (terrain-referenced navigation).

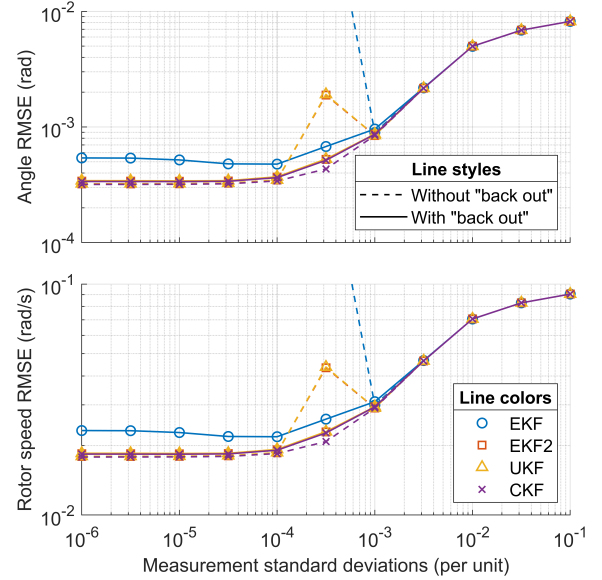


Fig. D.2. The root mean squared error of the state estimations with and without the “back out” step under different measurement noise setups (synchronous generator state estimation).

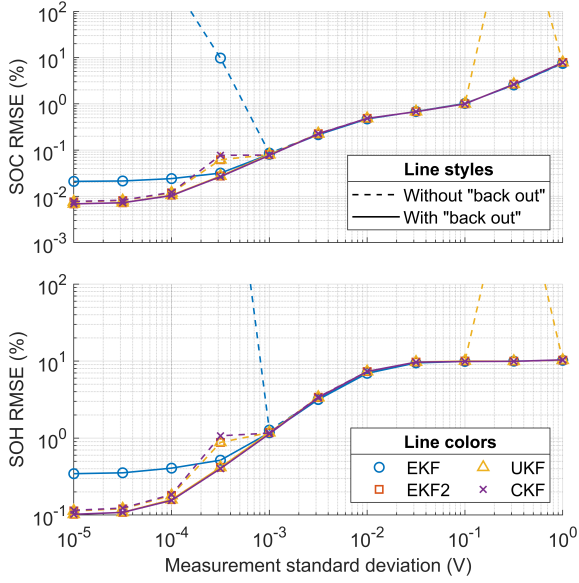


Fig. D.3. The root mean squared error of the state estimations with and without the “back out” step under different measurement noise setups (battery state estimation).

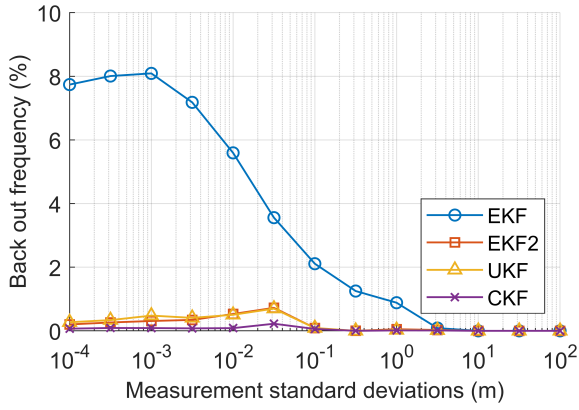


Fig. D.4. Percentage of state updates that trigger “back out” (3D target tracking).

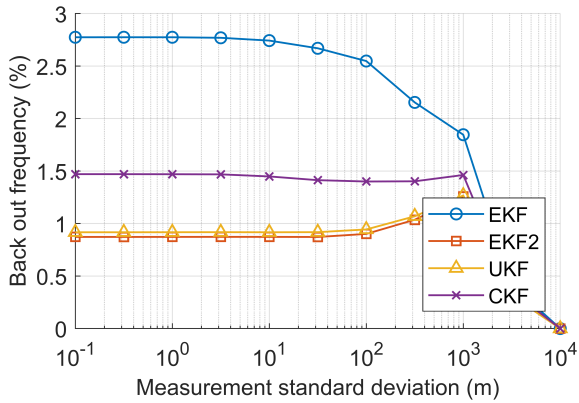


Fig. D.5. Percentage of state updates that trigger “back out” (terrain-referenced navigation).

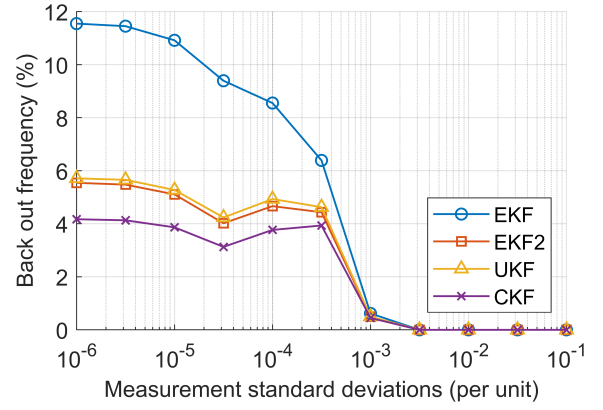


Fig. D.6. Percentage of state updates that trigger “back out” (synchronous generator state estimation).

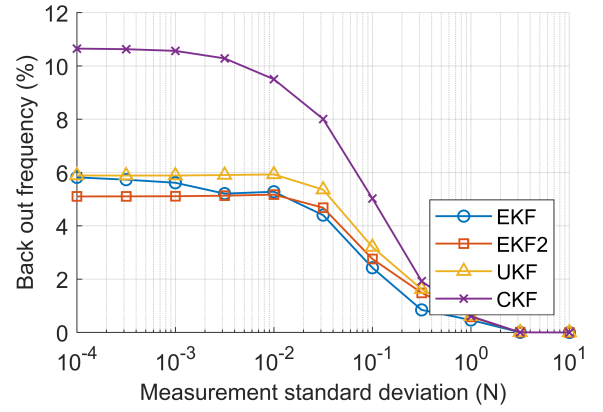


Fig. D.7. Percentage of state updates that trigger “back out” (pendulum state estimation).

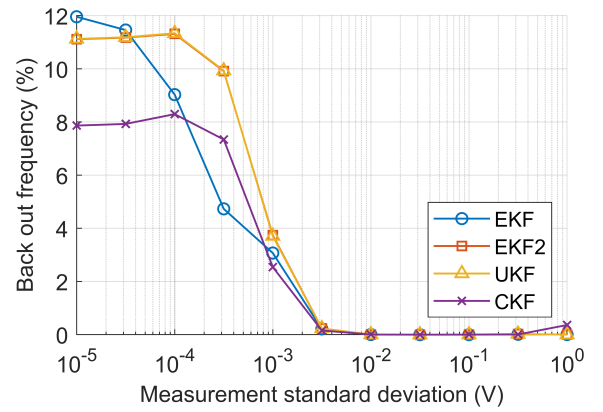


Fig. D.8. Percentage of state updates that trigger “back out” (battery state estimation).